MACHINE LEARNING · ROAD SAFETY RESEARCH

FINAL PROJECT

Car Crash Prediction on Busy Streets Using Logistic Regression and XGBoost

A comparative machine learning study of crash risk under urban traffic and weather conditions

PRESENTED BY

yashwant alli

01 · MOTIVATION

Busy intersections concentrate crash risk, yet most safety measures react only after collisions occur

The problem

Urban crashes cluster where traffic density, speed, and pedestrian flow interact with poor weather and road conditions.

Our objective

Estimate the probability of a crash from measurable traffic and environmental conditions, before it happens.

Why two models

Compare an interpretable linear model (Logistic Regression) against a nonlinear ensemble (XGBoost) on the same data.

A balanced dataset of 6,500 records and 10 features supports a fair model comparison

6,500
Total samples

10
Input features

5,200
Training set (80%)

1,300
Test set (20%)

Features captured per record

- Traffic density
- Rainfall level
- Pedestrian count
- signal
- Time of day
- Vehicle speed
- Visibility
- Road condition
- types of vehicles
- Intersections

Target variable

Binary outcome — **Crash** vs **No Crash** — with a near 50 / 50 class balance, so accuracy is a meaningful headline metric.

Split: stratified 80 / 20 train–test

Synthetic generation gives controlled, privacy-safe data with known risk relationships

Why synthetic data

Availability — granular, per-condition crash records are rarely public at the needed resolution.

Privacy — no real driver, vehicle, or location data is exposed.

Control — class balance and condition ranges are set deliberately for a clean comparison.

How relationships were encoded

Features were drawn from domain-informed distributions reflecting realistic urban traffic conditions.

Crash probability rises with rainfall, traffic density, and pedestrian volume, and falls in good road conditions and clear visibility.

Generated records were checked for plausibility before modelling.

A single preprocessing pipeline keeps both models on identical, comparable inputs

01

Encoding

Categorical fields (road condition, time of day) converted to numeric form.

02

Scaling

Features standardized so Logistic Regression coefficients stay comparable.

03

Stratified split

80 / 20 train–test split preserving the crash / no-crash balance.

04

Quality check

No missing values; feature correlations reviewed before training.

Both models are trained and evaluated on exactly the same processed features and the same test set, so any performance difference reflects the algorithm — not the data preparation.

Scaling applied for Logistic Regression; trees are scale-invariant

Logistic Regression gives a transparent, coefficient-level view of crash risk

A linear model combines the features into a single risk score, then maps it to a probability with the sigmoid function.

Each coefficient has a direct meaning: its sign shows whether a condition raises or lowers risk, and its exponential gives an odds ratio.

This interpretability makes it a strong, defensible baseline for a safety-critical setting.

RISK SCORE

$$\text{Risk} = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

PROBABILITY OF CRASH

$$P = 1 / (1 + e^{-\text{Risk}})$$

Output is bounded between 0 and 1 and read directly as a crash probability.

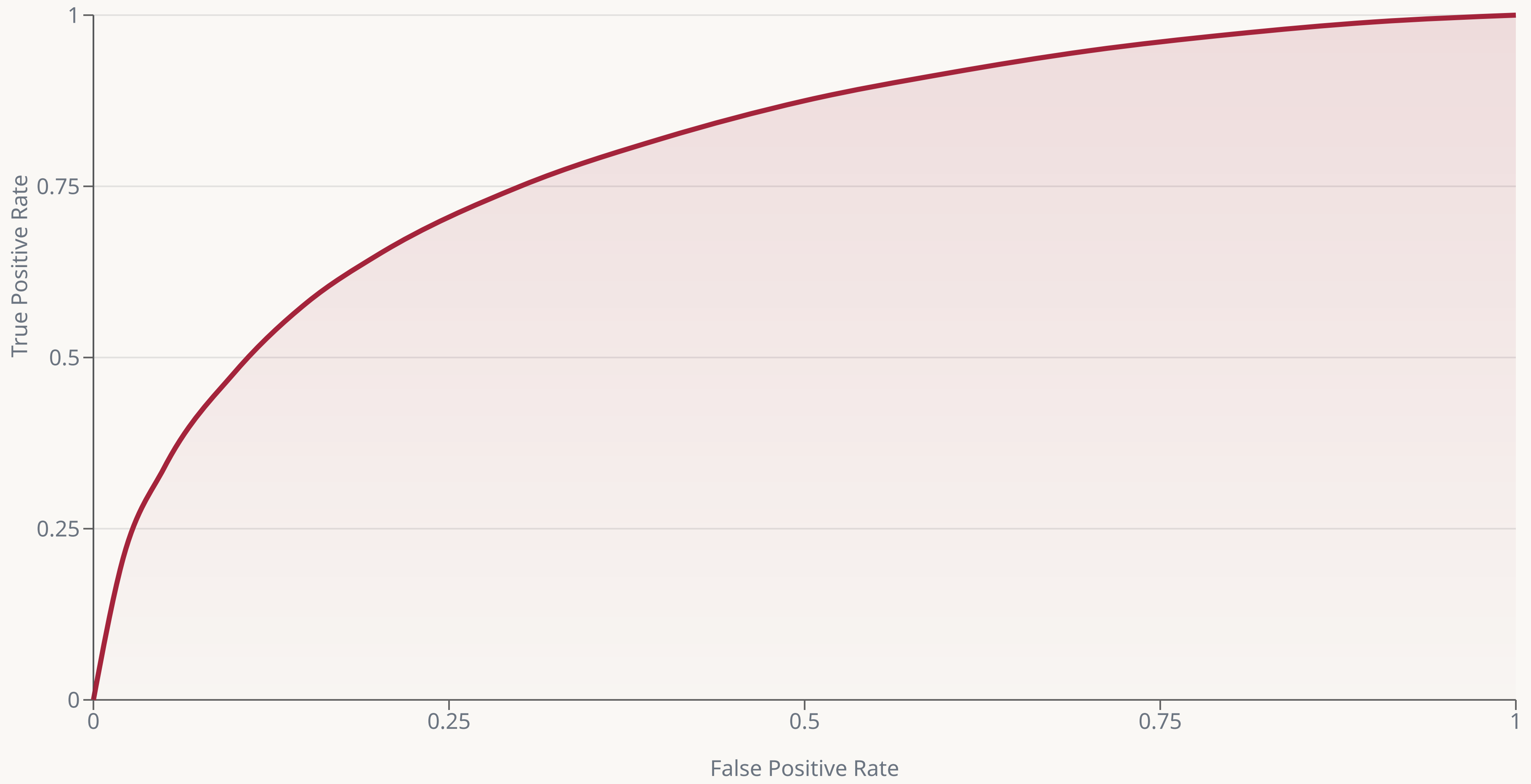
Solver: L2-regularized logistic regression on standardized features

Logistic Regression reaches 76.7% accuracy and a 0.854 ROC-AUC

0.767 Accuracy	**0.765** Precision
0.766 Recall	**0.766** F1-score

0.854 ROC-AUC

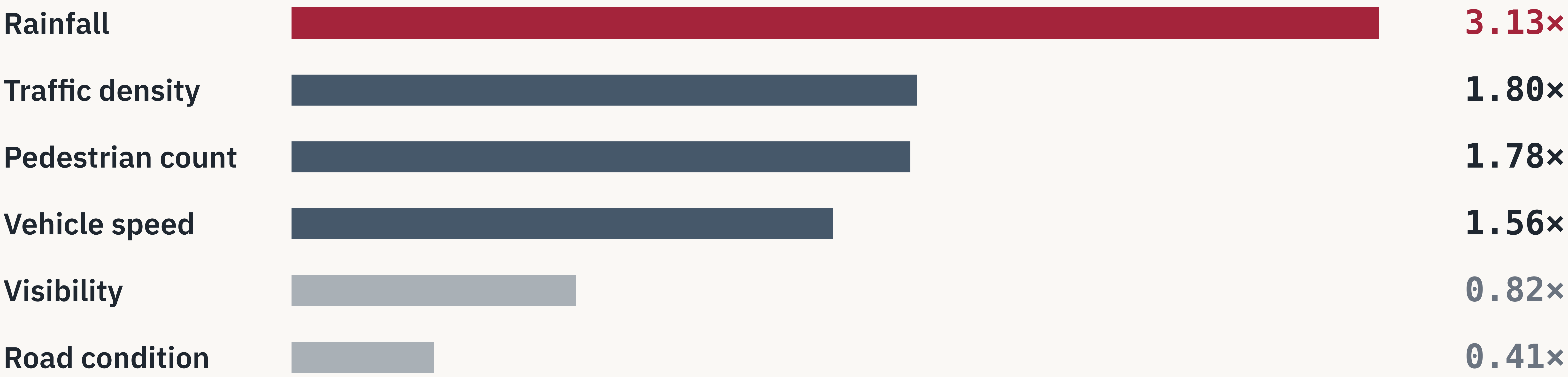
Receiver Operating Characteristic curve



Evaluated on the 1,300-record held-out test set · dashed line = random classifier

07 · MODEL A

Rainfall is the strongest risk driver — it more than triples the odds of a crash



Odds ratio > 1 raises crash odds. Wet weather, dense traffic, and high pedestrian volume are the dominant hazards.

Odds ratio < 1 is protective. Good road condition and clear visibility meaningfully reduce the odds of a crash.

Odds ratios = exp(coefficient) from the logistic model · bar length scaled to odds ratio

XGBoost adds nonlinear power by combining many regularized decision trees

Gradient boosting builds trees sequentially, with each new tree correcting the errors of the ensemble so far.

It captures nonlinear effects and feature interactions automatically — for example, how rainfall and speed compound risk together.

Built-in regularization controls model complexity and limits overfitting.

Sequential boosting

Weak learners added one at a time, each focused on the hardest cases.

Interaction-aware

Tree splits model combined conditions without manual feature engineering.

Regularized ensemble

Shrinkage and tree constraints trade a little interpretability for flexibility.

Trained on the same processed features as the logistic model

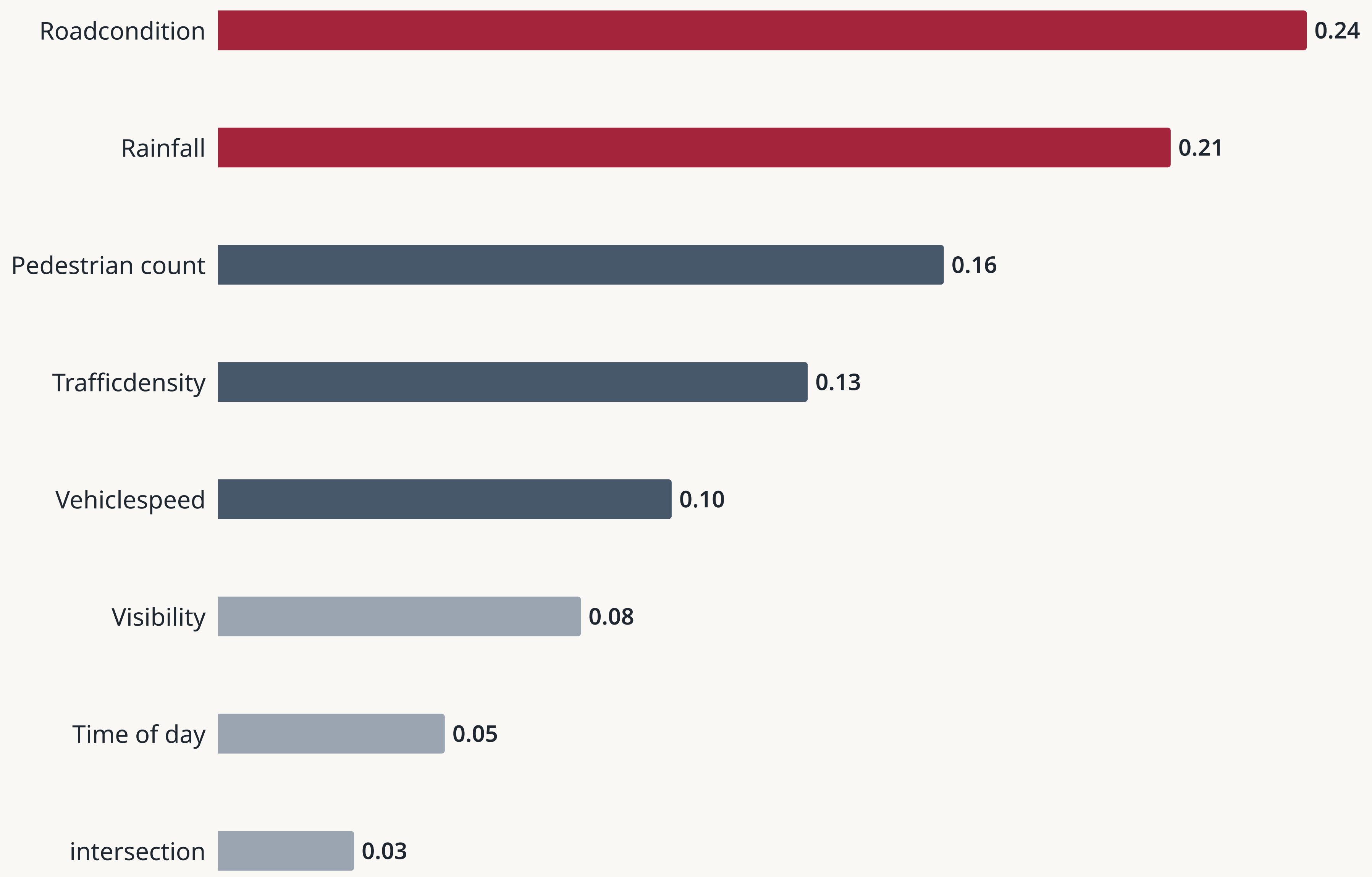
XGBoost matches the logistic model on accuracy, with a marginally lower 0.845 AUC

Accuracy	Precision	Recall	F1-score	ROC-AUC
0.765	0.765	0.760	0.762	0.845

On this dataset the extra flexibility of boosting does not translate into a performance gain. The two models are practically tied on accuracy, precision, recall, and F1, and the logistic model holds a small edge on AUC — a difference tested formally on slide 14.

Same 1,300-record held-out test set as the logistic model

XGBoost ranks road condition and rainfall as the most predictive features

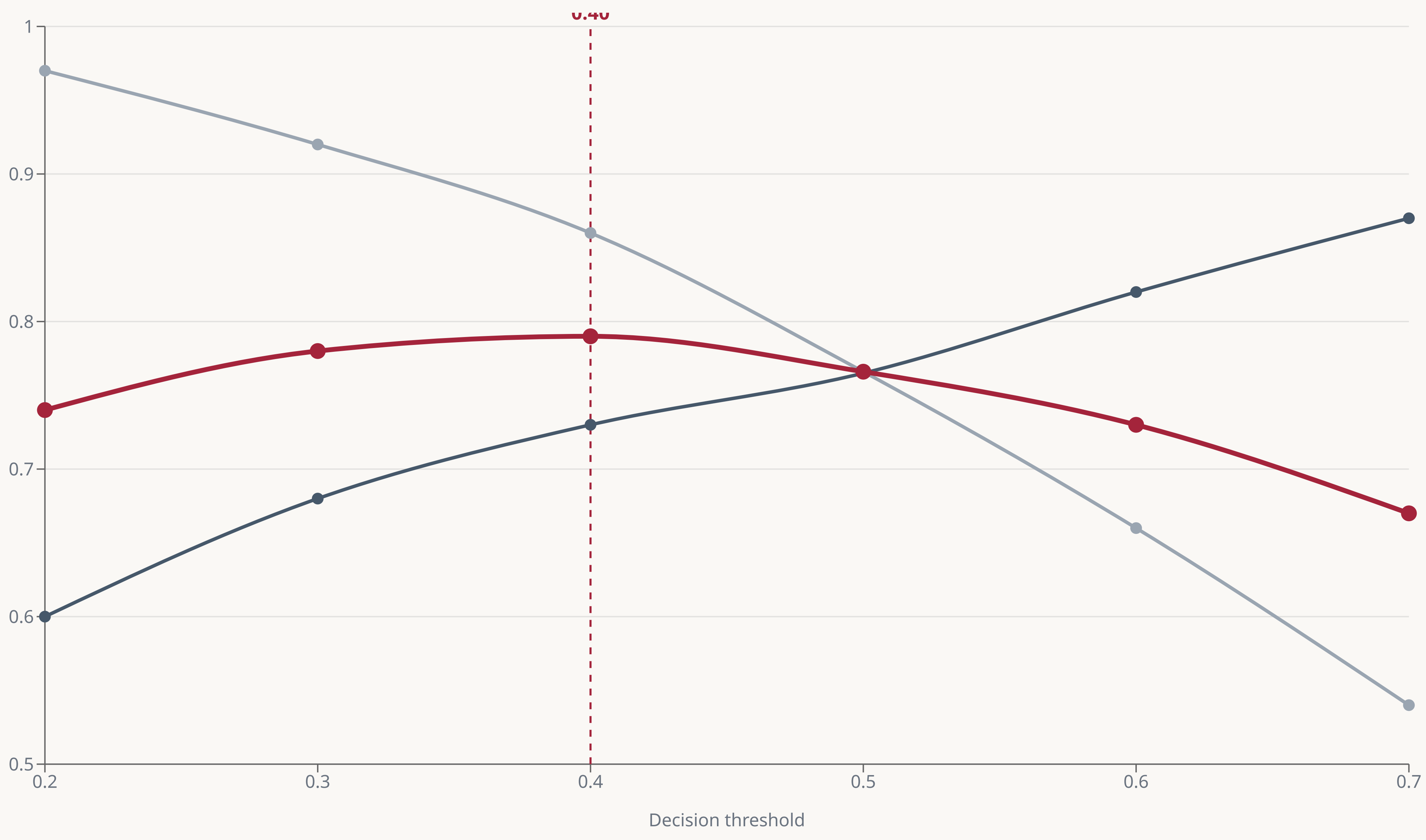


The tree model and the logistic coefficients largely agree: weather and road surface dominate, traffic and pedestrians follow.

This convergence across two very different algorithms strengthens confidence in which conditions actually drive crash risk.

Gain-based feature importance, normalized to sum to 1

Lowering the decision threshold to 0.40 maximizes F1 and recovers more true crashes



F1-score

Precision

Recall

At the default 0.50, precision and recall sit near 0.77. Shifting to **0.40** lifts recall to about 0.86 for a small precision cost — the right trade when missing a crash is worse than a false alarm.

Logistic Regression scores on the held-out test set across decision thresholds

The models are near-identical on every metric, with Logistic Regression ahead on AUC

Metric	Logistic Regression	XGBoost
Accuracy	0.767	0.765
Precision	0.765	0.765
Recall	0.766	0.760
F1-score	0.766	0.762
ROC-AUC	**0.854**	**0.845**

Accuracy, precision, recall, and F1 differ by at most half a percentage point. The only consistent separation is the **0.009 AUC gap** favouring Logistic Regression — small, but worth testing for significance.

Default 0.50 threshold · 1,300-record held-out test set

DeLong's test confirms the AUC gap is statistically significant ($p = 0.027$)

2.21
Z-statistic

0.027
p-value

Significance level

$\alpha = 0.05 \rightarrow p < \alpha$, reject H_0

DeLong's test compares two ROC-AUCs computed on the **same test set**, accounting for the correlation between the models' predictions.

The null hypothesis — that both models have equal AUC — is rejected at the 5% level.

Conclusion: Logistic Regression's 0.854 AUC is significantly higher than XGBoost's 0.845 on this dataset — the difference is unlikely to be due to chance.

DeLong et al. (1988) test for two correlated ROC curves · full formula in backup D

Logistic Regression is the recommended model — equal accuracy, higher AUC, full interpretability

01 Both models predict crashes at roughly **76–77% accuracy** on balanced data.

02 Logistic Regression's AUC edge is **statistically significant** (DeLong p = 0.027).

03 **Rainfall** is the dominant risk factor, followed by traffic density and pedestrian volume.

04 A **0.40 threshold** is recommended to prioritize catching real crashes.

For deployment, the interpretable model wins: it performs at least as well as the ensemble and tells operators why risk is elevated — essential for acting on it.

Results are bounded by synthetic data and a compact feature set

Synthetic foundation

Generated data encodes assumed relationships; it may not capture the noise, rare events, and complexity of real traffic.

Limited features

Ten features omit factors such as driver behaviour, vehicle type, signal timing, and intersection geometry.

No temporal modelling

Each record is treated independently; sequence and time-of-day dynamics are not modelled explicitly.

Unverified generalization

Performance on real-world crash data is untested; the 0.40 threshold should be re-tuned on field data.

Next steps move from synthetic proof-of-concept toward real-time deployment

01

Real-world validation

Retrain and test on field crash records to confirm the synthetic findings hold.

02

Richer features

Add spatial, temporal, and behavioural signals from sensors and traffic systems.

03

Advanced models

Explore sequence and deep-learning models where added complexity is justified.

04

Real-time scoring

Serve live risk estimates from streaming traffic and weather feeds.

05

System integration

Feed predictions into signal control and driver-warning infrastructure.

The goal: turn an interpretable model into an operational early-warning tool for busy streets.

References and acknowledgements

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Thank you

Questions and discussion welcome.

Confusion matrix — Logistic Regression on the test set

	Predicted: No Crash	Predicted: Crash
Actual: No Crash	502 True Negatives	152 False Positives
Actual: Crash	151 False Negatives	495 True Positives

Accuracy	0.767
Precision (crash)	0.765
Recall (crash)	0.766

Errors are roughly balanced: 152 false alarms against 151 missed crashes across the 1,300 test records.

Complete odds ratio table — Logistic Regression coefficients

Feature	Coefficient (β)	Odds ratio	Effect
Rainfall	+1.14	3.13	Strong risk increase
Traffic density	+0.59	1.80	Risk increase
Pedestrian count	+0.58	1.78	Risk increase
Vehicle speed	+0.44	1.56	Risk increase
Visibility	−0.20	0.82	Protective
Road condition (good)	−0.89	0.41	Strong protective

Odds ratio = exp(β). Values above 1 raise crash odds; values below 1 lower them. Coefficients are on standardized features, so magnitudes are directly comparable.

BACKUP C

Threshold sweep — precision, recall, and F1 by decision threshold

Threshold	Precision	Recall	F1-score
0.30	0.68	0.92	0.78
0.40	**0.73**	**0.86**	**0.79**
0.50	0.77	0.77	0.77

As the threshold falls, recall rises and precision drops. **0.40** gives the highest F1 (0.79) by balancing the two.

In a safety setting a missed crash costs more than a false alarm, so favouring recall at 0.40 over the default 0.50 is the appropriate operating point.

BACKUP D

DeLong's test — comparing two correlated ROC curves

TEST STATISTIC

$$Z = (AUC_1 - AUC_2) / \sqrt{(Var)}$$

VARIANCE TERM

$$Var = Var(AUC_1) + Var(AUC_2) - 2 \cdot Cov(AUC_1, AUC_2)$$

The covariance term accounts for both AUCs being estimated on the same test set.

DeLong's method estimates the variance and covariance of the AUCs non-parametrically, using the structural components of the Mann–Whitney statistic.

2.21
Computed Z

0.027
Two-sided p

With $|Z| = 2.21 > 1.96$, the difference is significant at $\alpha = 0.05$ — the logistic model's AUC is the higher of the two.